

A3 Report

Title of Issue: Battery Contamination

A3 Leader: Michael Hawk

Team: Kevin James, Randall Bates

CAR/PAR No:

Date Opened: 7/14/2025

Date Closed: 8/1/2025



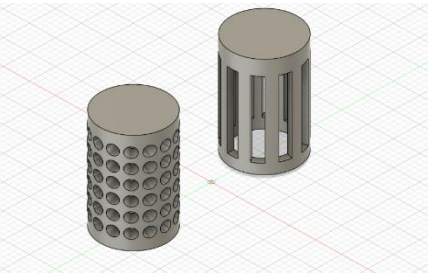
Step 1: Define the Problem: (What, When, Where, First time observed, Total population)

Clogged standing tubes causing contamination of batteries

Step 2: Problem Breakdown (Pictures, Drawings, Charts, etc. to illustrate problem as much as possible)

-Earplugs and other debris floating in charging tubs get caught in standing tubes, which leads to clogging and water rises above the batteries, causing contamination and scrap

-Because there is really no way to prevent debris from getting into the tubes, the best course of action is to work on the tubes themselves and develop a countermeasure



Step 3: Set the Goal/Target: (Define the desired situation once the problem is solved)

Reduce contaminated batteries by preventing the clogging of standing tubes

Step 4: Root Cause Analysis (Identify possible root causes and validate their contribution to the problem)

Possible Cause	Why? Because...	Proposed Countermeasure to be implemented
Batteries contaminated by water flooding the batteries	Why? Objects get clogged in the stand tubes and water can't flow	Unable to keep operators accountable for the debris so the best option is to not allow the stand tubes to be cloggable. Create new stand tube design that allows water flow even if clogged.
	Why? Earplugs, gloves, etc. fall or get placed in the tubs	
	Why? Operators may drop them or forget they set them down	
	Why?	
	Why?	Responsible: Full Team

Step 5: Develop and Prioritize Countermeasures

Prioritize implementation of countermeasures by using the calculated value in Overall column = Cost score X Feasibility score X Effectiveness score. Prioritize Higher Overall First								
Score 3 = Low Cost, Easy to Implement, Effective Score 2 = Mid \$\$, long time to Implement, Some what Effective Score 1 = High \$\$\$, Hard to Implement, Low Effectiveness								
Cause No.	Root Cause	Countermeasure	Cost \$	X	Feasibility X	Effectiveness	= Overall	Comments
1	Clogged Standing Tubes	Develop new standing tubes to prevent clogging	2		3	2	12	
2								
3								

Step 6: Implement Countermeasures

ID	Countermeasure Implementation Task	Start	Finish	Responsible	Weeks after start date (Check the boxes for duration to complete task)												
					1	2	3	4	5	6	7	8	9	10	11	12	
1	Design Prototypes	7/17/2025	7/18/2025	Michael H.													
2	Create Prototypes	7/21/2025	7/28/2025	Fabrication													
3	Test Prototypes	7/22/2025	8/1/2025	Full Team													
4																	
5																	
6																	

Step 7: Countermeasure Effectiveness (Validate using data)

Set the Goal+limits	Weeks after start date (Trend the performance every week)											
	1	2	3	4	5	6	7	8	9	10	11	12
	Actual:											
- Validate Countermeasure(s) had the expected impact on the problem defined in Step 1. - Trend the actual levels weekly and compare to the Goal/Target defined in Step 3.												

Step 8a: Standardize Successes (If a question answer is "No", the appropriate task needs to be completed to answer "Yes")

No	Process Changes	Yes	No	N/A	No	Tools / Equipment Change	Yes	No	N/A
1	Changes are reflected in the Work Instructions	☐	☐	☒	1	Drawings for Tools / Equipment reflect the changes	☐	☐	☒
2	The Issue will not occur when following the work instruction	☐	☐	☒	2	Operators trained on the use of new tools/equipment	☐	☐	☒
3	The operator can identify/detect the issue	☐	☐	☒	3	Operators trained on Autonomous Maintenance	☐	☐	☒
4	Changes are reflected in the control plan	☐	☐	☒	4	Changes reflected on the Preventive Maintenance	☐	☐	☒
5	Changes are reflected in the PFMEA	☐	☐	☒	Other Forms and Next Steps:				
6	Operators/Leads/Supervisors trained in new process	☐	☐	☒					

Step 8b: Lessons Learned and Shareability (Actual Outcome)

Implementation Contact and Information	Can this countermeasures be implemented in similar processes in the plant or in other plants? Which ones?	This countermeasure can be implemented
Plant Contact:	Lessons learned / Outcome:	
Phone number:	The main goal was to find a way to reduce battery contamination and although the clogging is an issue, it is not the root cause. The new stand tubes should reduce the amount of scrap, but not eliminate it completely. It would be beneficial to continue to improve the process until scrap is eliminated completely.	
Cost per line/area:		
Vendor / Supplier:		
Drawing / ECO:		